

WELCOME TO OPTIMIZATION MODELS AND METHODS!

Throughout this course, we are going to learn how to use Gurobi Optimization Software - the fastest solver!

Before we dive in to modeling and solving problems, let's play an educational game to see why optimization is **valuable** and **important**, why it's **difficult**, and why solvers and other optimization algorithms are essential in finding an optimal solution.

Burrito Optimization Game

HOW TO PLAY?

Guroble needs your assistance to plan where to place its burrito trucks to serve hungry customers throughout the city and maximize its profit. Truck placement has to be carefully planned, because every truck has a cost, but its ability to earn revenue depends on how close it is to potential customers. On some days, special events increase the demand in some regions of the city; on other days, the weather affects customers' willingness to walk to a burrito truck. Help Guroble succeed by placing the burrito trucks! Gurobi Optimization Software will compare your results to the optimal solution and will rank you accordingly!

READY TO COMPETE FOR AN ACTUAL BURRITO?

1. Go to this link: <https://www.gurobi.com/burrito-optimization-game/>
2. Click **"Register to Play"**
3. Registration should not take more than 3 minutes! **Use your Columbia e-mail address** so that you can get the free academic license
4. After registering, you should be good to go - click **"Championship Mode"** button to compete against your classmates!
5. Enter the match code as: **[will be provided during the lecture]**
6. Enter your display name
7. Click **"Join Match"** and you are in!



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